Surveillance: SURV controls, ECAM SURV lines and TCAS

The A380X SURV pages and pedestal SURV panel work and stay in step, the ECAM procedures sense the SURV system selection, and the TCAS of both aircraft shows TAs while RAs are inhibited, with the real low-altitude hysteresis, plus the FCOM TCAS messages, THRT display and TRAF / SYS pushbutton fixes

Aircraft	A380X (SURV panel, VD messages, ECAM lines), A32NX + A380X (TCAS), A32NX (GPWS / TAWS)
Where to find it	A380X MFD **SURV** pages and pedestal SURV panel; A380X VD and E/WD; ND traffic display and TCAS controls of both aircraft
Branch	`core/surveillance`, based on `master` (FlyByWire master 76db0aef1 merged in); independent. `core/a380x-mfd-fms` is stacked on it (it adds the FCOM layout of the SURV pages and the SURV memos).
Status	SURV panel and MFD SURV wiring tested in the sim by the user; VD messages and ECAM SURV lines tested in the sim (September 2026); the three TCAS changes tested in the sim by the user: OK. (TCAS display, ND messages, TRAF pb, SYS pbs; added): not sim-tested, unit tests only.
Pictures	Captain MFD, ND, VD and E/WD in the local browser harness, running the full development build installed in the sim (it also has the FCOM layout of the SURV pages from `core/a380x-mfd-fms`, so the SURV page frames on this branch alone look slightly different); the SURV CONTROLS in-sim screenshot; the TCAS symbol legend is a drawing (the harness has no traffic); the A32NX ND TERR AHEAD pictures are in-sim screenshots.

1. A380X SURV controls, STATUS & SWITCHING and the pedestal SURV panel
2. A380X VD: separate NO TERR / NO WX DATA messages
3. A380X ECAM procedures: WXR & TAWS and XPDR & TCAS lines sensed
4. A380X ECAM LINE-UP checklist: CABIN CREW line
5. TCAS: intruders shown and announced as TAs while RAs are inhibited
6. TCAS: low-altitude RA and TA aural inhibitions with the real hysteresis
7. A32NX: TCAS message at the left of the bottom line of the ND
8. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)
9. TCAS traffic display, ND TCAS messages, TRAF pb, WXR/TAWS SYS pbs

1. A380X SURV controls, STATUS & SWITCHING and the pedestal SURV panel

> **Aircraft:** A380X | **Where:** MFD SURV pages; pedestal SURV panel | **Status:** Tested in the sim by the user: OK (pedestal buttons and MFD pages switch in step; WXR buttons drive the radar).

What it is for

The weather radar block of the MFD SURV CONTROLS page, the STATUS & SWITCHING system selection and the pedestal SURV panel buttons were stubs. They now work and drive the same systems.

What it is for

The A380 has no separate radar, TCAS or TAWS control panels: they are on the MFD SURV pages and the pedestal SURV panel (FCOM DSC-34-20-60-50). In FBW the weather radar block of the page was a stub (every button greyed), STATUS & SWITCHING did not reflect failures, and no button of the pedestal panel worked. Now they do, and the page and the panel stay in step.

SURV CONTROLS

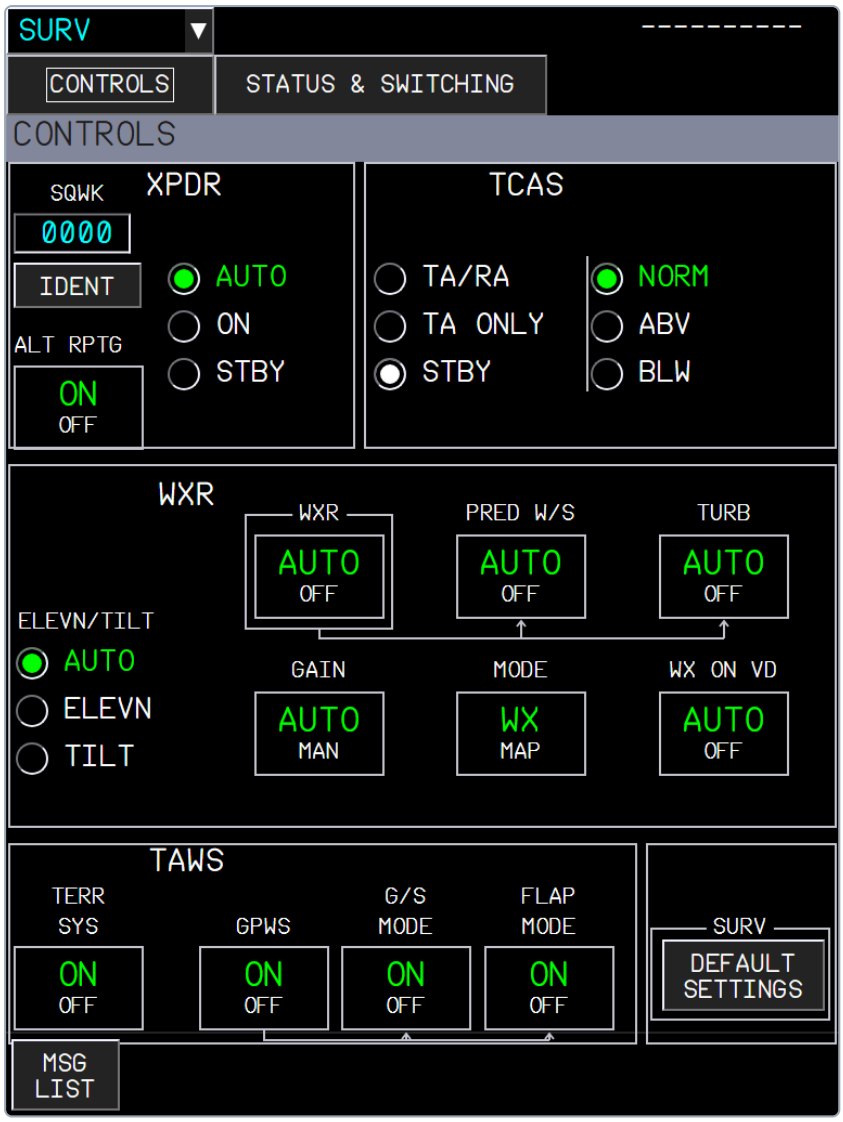


Figure 1. SURV CONTROLS: XPDR (code, IDENT, AUTO / ON / STBY, ALT RPTG), TCAS (TA/RA, TA ONLY, STBY; NORM, ABV, BLW), WXR, TAWS, and DEFAULT SETTINGS.

WXR button	Effect
WXR AUTO / OFF	Weather radar on or off (the ND radar picture of `core/nd-weather-terrain`).
PRED W/S AUTO / OFF	Predictive windshear (written for the systems; MSFS has no predictive windshear).
TURB AUTO / OFF	Turbulence detection (magenta on the ND with the native radar of `core/nd-weather-terrain`).
GAIN AUTO / MAN	Radar gain (written; MSFS has no gain).
MODE WX / MAP	Weather or ground mapping.
WX ON VD AUTO / OFF	Weather on the vertical display.
ELEVN / TILT	AUTO, or a manual elevation or tilt.

The WXR buttons are inhibited only when the selected WXR system has really failed.

STATUS & SWITCHING

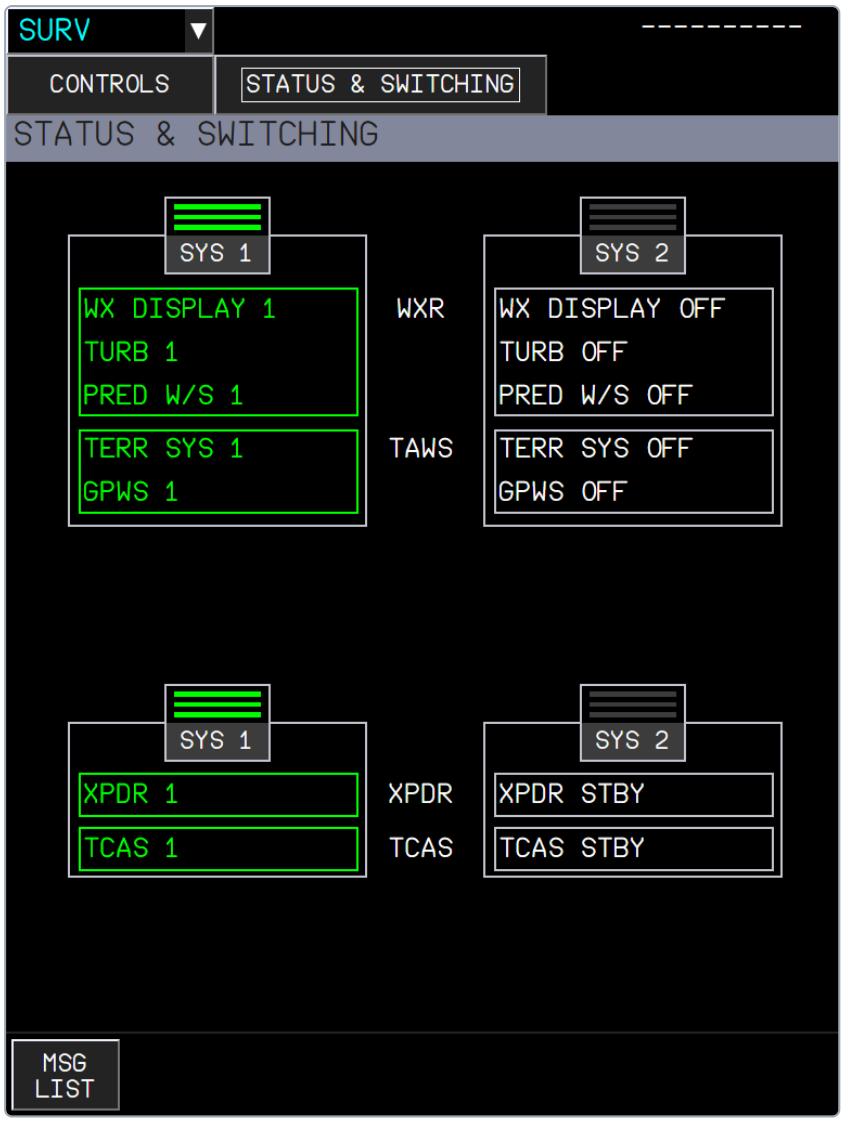


Figure 2. STATUS & SWITCHING: SYS 1 / SYS 2 of the WXR and TAWS (top) and of the XPDR and TCAS (bottom); the selected system is lit, each item shows its real state.

- WXR DISPLAY, TURB, PRED W/S, TERR SYS, GPWS items show the real failure of each system.
- XPDR / TCAS SYS 1 / SYS 2 select the transponder system (the one the TCAS uses) and light from it.



Figure 3. The SURV CONTROLS page in the sim (in-sim screenshot).

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The pedestal SURV panel

The panel buttons never worked: their light template runs the button code only when the panel is powered, and the pedestal never told it so. They are powered from DC 1 now and linked to the MFD pages:

Panel button	Does	Same as
WXR TAWS SYS 1 / SYS 2	Selects the WXR / TAWS system	STATUS & SWITCHING, top
XPDR TCAS SYS 1 / SYS 2	Selects the transponder system	STATUS & SWITCHING, bottom
G/S MODE	Glide slope mode of the GPWS on / off	SURV CONTROLS, TAWS
TCAS ABV / BLW / TA ONLY	TCAS display and mode	SURV CONTROLS, TCAS

Each button lights from the same value the page uses, so pressing one on the pedestal changes the page, and the reverse.

Limitations and why

Limitation	Why	What to do
PRED W/S and GAIN MAN do nothing visible.	The MSFS radar has no predictive windshear and no gain; the values are written for when it does.	-
One MSFS radar for both systems.	SYS 1 / SYS 2 select the failure state and the lights; the picture comes from the same sim radar.	-

> **NOTE**
> Code clean-up on this branch with no visible effect: the ECAM procedure tables read the selected XPDR & TCAS system through one helper.

How to use

1. Open the SURV pages: MFD system menu **SURV** (or the KCCU SURV key), then **CONTROLS** or **STATUS & SWITCHING**.
2. On CONTROLS set the WXR (WXR, PRED W/S, TURB, GAIN, MODE, WX ON VD, ELEVN/TILT), the TCAS (TA/RA, TA ONLY, STBY; NORM, ABV, BLW) and the TAWS options.
3. On STATUS & SWITCHING, or with the pedestal buttons, select WXR / TAWS SYS 1 or SYS 2 and XPDR / TCAS SYS 1 or SYS 2.
4. Use the pedestal TCAS ABV / BLW / TA ONLY and G/S MODE buttons as shortcuts: the MFD page follows.

How to test

#	Do this	Expected result
1	A380X on the ground, aircraft powered (DC 1 bus powered). MFD > SURV > **STATUS & SWITCHING**.	The selected WXR / TAWS and XPDR / TCAS systems are lit; each item shows the real state of its system.
2	Press **XPDR TCAS SYS 2** on the pedestal SURV panel.	The pedestal button lights; on the MFD page XPDR / TCAS SYS 2 is now the selected one.
3	Select XPDR / TCAS **SYS 1** on the MFD page.	The pedestal SYS 1 button lights, SYS 2 goes off.
4	Press **WXR TAWS SYS 2** on the pedestal, then SYS 1.	The WXR / TAWS selection on the MFD page follows each press.
5	MFD > SURV > **CONTROLS**. Press the pedestal **TA ONLY** button.	The SURV CONTROLS TCAS block shows TA ONLY and the pedestal button is lit; the ND shows TA ONLY.
6	Press the pedestal **ABV**, then **BLW** button.	The TCAS NORM / ABV / BLW selection on the page follows; selecting NORM on the page turns the pedestal light off.
7	Press the pedestal **G/S MODE** button.	The TAWS G/S MODE on SURV CONTROLS changes with it (and back on the second press).
8	On SURV CONTROLS set **WXR** OFF, then AUTO (needs the native radar of `core/nd-weather-terrain` for the ND picture).	The ND radar picture goes away and comes back; the WXR buttons are greyed only if the selected WXR system has failed.
9	Cut the DC 1 bus (or start cold and dark) and press a pedestal SURV button.	Nothing happens and the buttons stay dark.

2. A380X VD: separate NO TERR / NO WX DATA messages

> **Aircraft:** A380X | **Where:** Vertical display (VD) under the ND | **Status:** Tested in the sim (September 2026).

What it is for

TERR SYS OFF removed both the terrain and the weather from the VD message logic, so the VD always said "NO TERR AND WX DATA AVAILABLE". Each source now has its own message.

Messages on the vertical display

The VD message was always "NO TERR AND WX DATA AVAILABLE" when TERR SYS was off. Now:

Situation	VD message
TERR SYS OFF only	NO TERR DATA AVAILABLE
WXR failed or OFF, or WX ON VD OFF	NO WX DATA AVAILABLE
Both	NO TERR AND WX DATA AVAILABLE

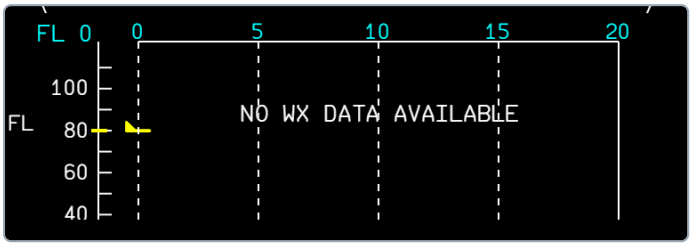


Figure 4. WXR off: NO WX DATA AVAILABLE.

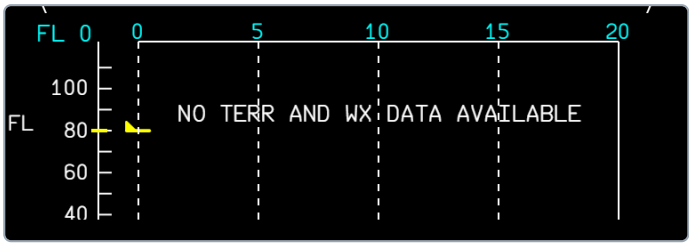


Figure 5. TERR SYS off and WXR off: NO TERR AND WX DATA AVAILABLE.

How to use

1. Nothing to set: the VD message follows the TERR SYS, WXR and WX ON VD buttons of MFD > SURV > CONTROLS.

How to test

#	Do this	Expected result
1	A380X, ND in ARC or ROSE NAV so that the VD shows. MFD > SURV > CONTROLS: TERR SYS, WXR and WX ON VD all on.	No "NO ... DATA" message on the VD.
2	Set **TERR SYS** OFF only.	VD: **NO TERR DATA AVAILABLE**.
3	TERR SYS back ON, set **WX ON VD** OFF.	VD: **NO WX DATA AVAILABLE**.
4	WX ON VD back ON, set **WXR** OFF.	VD: **NO WX DATA AVAILABLE**.
5	With WXR OFF, also set TERR SYS OFF.	VD: **NO TERR AND WX DATA AVAILABLE**.

3. A380X ECAM procedures: WXR & TAWS and XPDR & TCAS lines sensed

> **Aircraft:** A380X | **Where:** E/WD procedures; MFD SURV STATUS & SWITCHING | **Status:** Tested in the sim (September 2026).

What it is for

The dual ADR and IR fault procedures ask the crew to move the WXR & TAWS and the XPDR & TCAS to the system that still has valid data. Those lines were hard-coded as done, so the ECAM ticked them even when the wrong system stayed selected. They now follow the selection made on the MFD SURV STATUS & SWITCHING page, and the TCAS 1 / 2 FAULT procedures show their XPDR & TCAS line, which was hidden.

Where the systems are selected

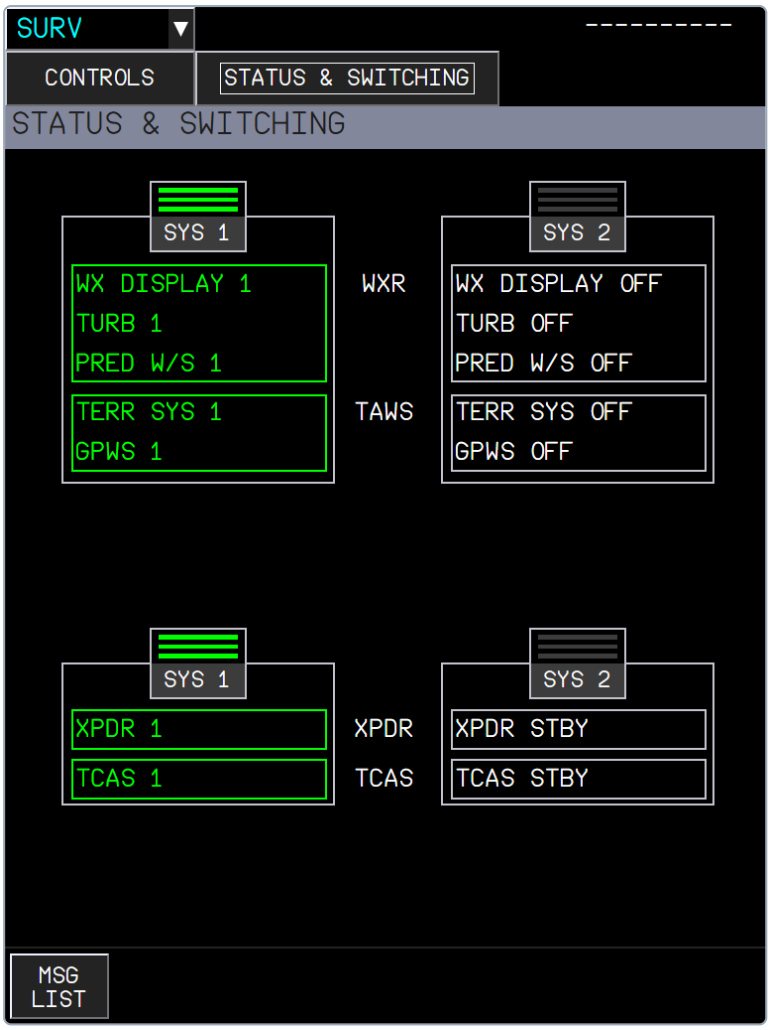


Figure 6. MFD SURV STATUS & SWITCHING: the SYS 1 / SYS 2 buttons of WXR & TAWS (top) and XPDR & TCAS (bottom).

What you see

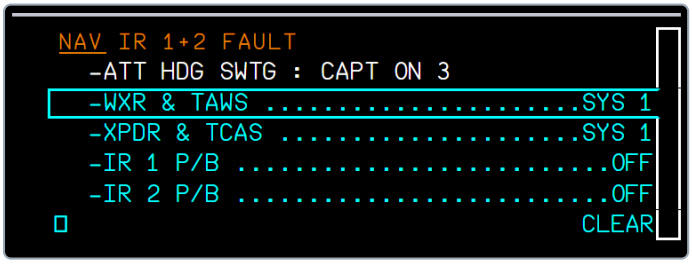


Figure 7. IR 1+2 FAULT with both systems on SYS 2: the two lines wait, in cyan.

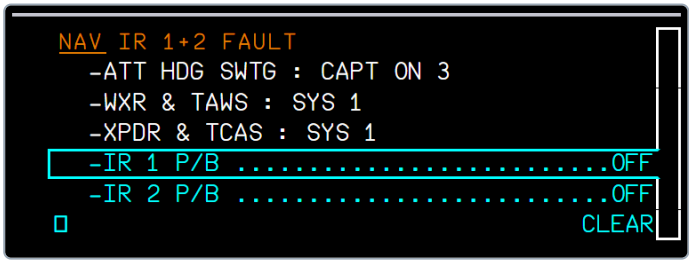


Figure 8. After selecting SYS 1 on the SURV page: the lines are done (white) and the cursor moves on.

Procedure	WXR & TAWS	XPDR & TCAS
ADR 1+2 FAULT, ADR 2+3 FAULT, IR 1+2 FAULT, IR 2+3 FAULT	SYS 1	SYS 1
ADR 1+3 FAULT, IR 1+3 FAULT	SYS 2	SYS 2
TCAS 1 FAULT	-	SYS 2
TCAS 2 FAULT	-	SYS 1

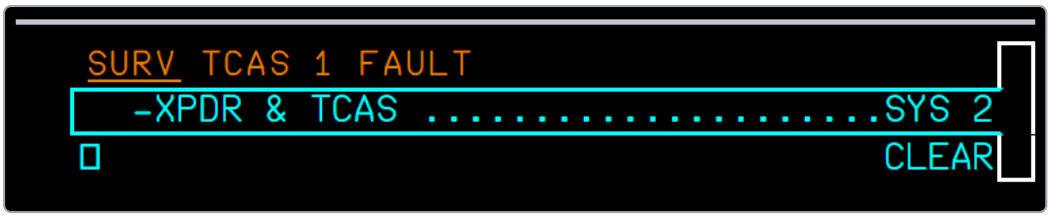


Figure 9. TCAS 1 FAULT: the XPDR & TCAS line is now shown.

Situation	What you should see
The right system is already selected when the fault appears	The line is shown already done.
The crew selects the other system	The line ticks within a second; the cursor moves to the next line.
The crew goes back to the failed side	The line becomes due again (it follows the live selection).

Limitations and why

Limitation	Why	What to do
TCAS 1 FAULT and TCAS 2 FAULT come together.	FBW has one TCAS computer: both procedures read the same TCAS fault (an FBW note in the code).	Select the system the procedure you follow asks for.

How to use

1. When an ADR / IR dual fault or a TCAS fault procedure asks for **WXR & TAWS ... SYS n** or **XPDR & TCAS ... SYS n**, select that system on MFD > SURV > STATUS & SWITCHING (or with the pedestal SYS buttons).
2. The line ticks by itself; the E/WD cursor moves to the next line.

How to test

#	Do this	Expected result
1	A380X on the ground, one engine running (the ADR faults are not raised in flight phases 1 and 12). MFD > SURV > STATUS & SWITCHING: select WXR / TAWS **SYS 2** and XPDR / TCAS **SYS 2**.	Both SYS 2 selected.
2	Overhead ADIRS panel: set the **ADR 1** and **ADR 2** pushbuttons OFF.	E/WD: **ADR 1+2 FAULT**. Its WXR & TAWS ... SYS 1 and XPDR & TCAS ... SYS 1 lines are cyan (to do).
3	Select WXR / TAWS **SYS 1** on the SURV page.	Within a second the WXR & TAWS line turns white (done).
4	Select XPDR / TCAS **SYS 1** (page or pedestal).	The XPDR & TCAS line is done; the cursor moves on.
5	Select XPDR / TCAS SYS 2 again.	The XPDR & TCAS line is due again (it follows the live selection). Set the ADR pushbuttons back ON afterwards.
6	Raise a TCAS fault: flyPad > Failures > ATA 34 **RA SYS A** and **RA SYS B** (the TCAS uses radio altimeter 1, else 2), on the ground.	E/WD: TCAS 1 FAULT and TCAS 2 FAULT, each now with its XPDR & TCAS line (before: hidden). The radio altimeter alerts show too.
7	Select XPDR / TCAS SYS 2, then SYS 1.	With SYS 2 the TCAS 1 FAULT line is done, with SYS 1 the TCAS 2 FAULT line. Clear the failures afterwards.

4. A380X ECAM LINE-UP checklist: CABIN CREW line

> **Aircraft:** A380X | **Where:** E/WD normal checklist LINE-UP | **Status:** Tested in the sim (September 2026).

What it is for

The A380 FCOM BEFORE TAKEOFF checklist has a CABIN CREW item; the ECAM LINE-UP checklist did not.

LINE-UP checklist

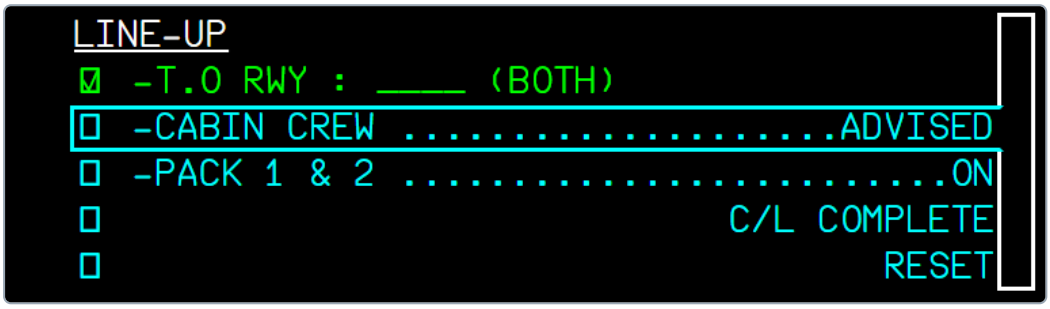


Figure 10. ECAM LINE-UP: the CABIN CREW ... ADVISED line (crew check).

The ECAM LINE-UP checklist gets the CABIN CREW line that the flyPad checklist already had (A380 FCOM BEFORE TAKEOFF checklist: CABIN CREW ... ADVISE). It is ticked by the crew.

Limitations and why

Limitation	Why	What to do
No SURV lines in the normal checklists.	The A380 FCOM normal checklists (PRO-NOR-C-L) have none (SURV, WX & TERR and TCAS are SOP flow items); an earlier version of this change added some and was reverted.	-

How to use

1. At line-up, call the ECAM LINE-UP checklist (C/L key on the ECAM control panel).
2. Advise the cabin crew, then tick the **CABIN CREW ... ADVISED** line.

How to test

#	Do this	Expected result
1	A380X ready for takeoff. Press **C/L** on the ECAM control panel and go to the **LINE-UP** checklist.	The checklist shows a **CABIN CREW ... ADVISED** line, in cyan (to do).
2	Tick the line (ECAM control panel or KCCU).	The line turns white (done); it is not ticked by itself.

5. TCAS: intruders shown and announced as TAs while RAs are inhibited

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display; TCAS mode | **Status:** Tested in the sim by the user: OK.

What it is for

With every RA inhibited (TA ONLY selected, or low: below 900 ft radio altitude in descent, until 1100 ft in climb, see the next feature), the TCAS kept classifying an intruder that met the RA criteria as an RA. The ND drew it as a red RA square, and since it was not a TA, there was no "TRAFFIC, TRAFFIC" and no TA state. A real TCAS in TA ONLY shows it as an amber TA with the TA aural. The classification is now capped at TA while all RAs are inhibited.



Figure 11. The ND traffic symbols: other, proximate, TA, RA, with the relative altitude in hundreds of feet and the vertical trend arrow.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

What you see



Figure 12. A32NX: TA ONLY, centred at the bottom of the ND (A320 FCOM DSC-34-SURV-60-20 figure).

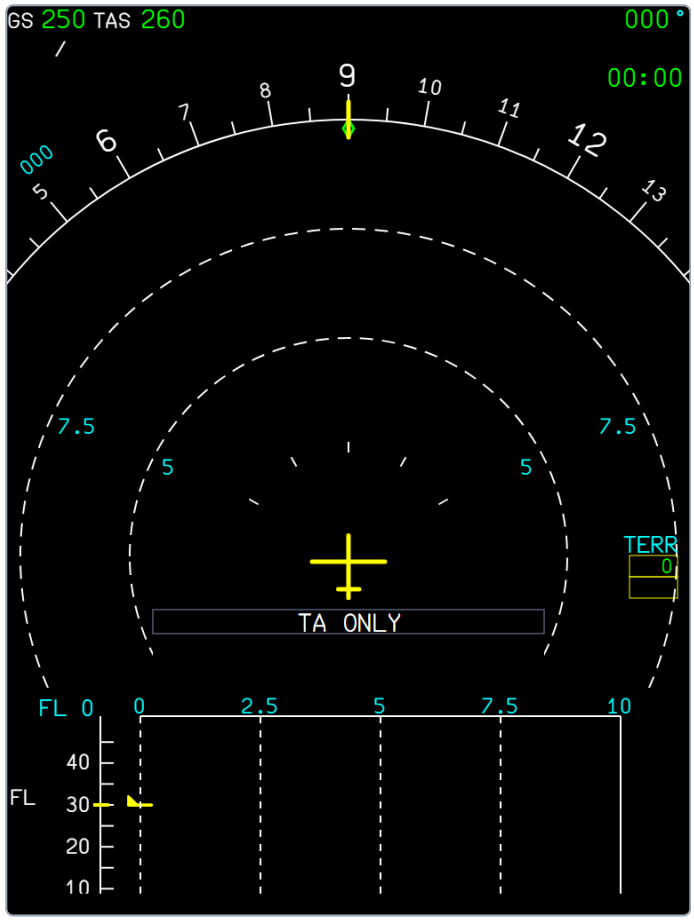


Figure 13. A380X: TA ONLY in the message box.

Situation	What you should see
TA ONLY selected, intruder meeting the RA criteria	Amber TA circle and "TRAFFIC, TRAFFIC" (before: red RA square, no aural).
Below 900 ft radio altitude in descent, until 1100 ft in climb (all RAs inhibited)	Same: TAs only, TA ONLY on the ND.
Partial inhibits (no CLIMB / DESCEND / INCREASE DESCENT RA)	Unchanged: RAs are still issued.
An RA already running when the inhibit starts	It runs to its own end.

Limitations and why

Limitation	Why	What to do

Same change on both aircraft.	The A32NX TcasComputer and the A380X LegacyTcasComputer share the classification code.	-

How to use

1. Select TA ONLY: A32NX ATC/TCAS panel TCAS mode selector to **TA**; A380X MFD > SURV > CONTROLS TCAS **TA ONLY** (or the pedestal TA ONLY button).
2. Traffic that would give an RA is shown as an amber TA with the "TRAFFIC, TRAFFIC" aural.

How to test

#	Do this	Expected result
1	In flight above 1100 ft radio altitude with traffic (AI or multiplayer) converging on you. TCAS in TA/RA.	An intruder meeting the RA criteria is a red square with an RA aural.
2	Select TA ONLY (A32NX: TCAS mode selector TA; A380X: SURV CONTROLS TA ONLY).	TA ONLY on the ND. The same intruder is now an amber TA circle with "TRAFFIC, TRAFFIC" (before: a red RA square and no aural).
3	Back to TA/RA during an RA already running, then TA ONLY again.	An RA already running when the inhibition starts runs to its own end.

6. TCAS: low-altitude RA and TA aural inhibitions with the real hysteresis

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display (TA ONLY message); TCAS aurals | **Status:** Tested in the sim by the user: OK (with the TA ONLY change).

What it is for

The TCAS inhibits all RAs close to the ground, and lower down the TA aural too. Both used a flat threshold (1000 ft and 500 ft radio altitude). The A380 FCTM (Supplementary Information, TCAS) gives a hysteresis, the FCOM "below 1000 ft ±100 ft". Both TCAS computers now use it:

Inhibition	Starts (descending through)	Ends (climbing through)
All RAs (TA ONLY on the ND)	900 ft radio altitude	1100 ft radio altitude
TA aural ("TRAFFIC, TRAFFIC")	400 ft radio altitude	600 ft radio altitude

- Without a valid radio altitude, neither low-altitude inhibition applies.
- The four altitudes are named constants of the TCAS code (no other effect).

How to use

1. Nothing to set: the inhibitions follow the radio altitude.

How to test

#	Do this	Expected result
1	A32NX or A380X on approach, TCAS in TA/RA, descending through 1000 ft radio altitude. Watch the ND.	No TA ONLY message yet at 1000 ft.
2	Continue below 900 ft radio altitude.	**TA ONLY** appears on the ND (all RAs inhibited).
3	Go around and climb.	TA ONLY stays up to 1100 ft radio altitude and disappears above it.

4	With traffic close by on a low approach (AI or multiplayer), descend below 400 ft radio altitude.	The TA symbol stays, the "TRAFFIC, TRAFFIC" aural is not given; it is given again above 600 ft in the climb.

7. A32NX: TCAS message at the left of the bottom line of the ND

> **Aircraft:** A32NX | **Where:** ND, bottom line, left of centre | **Status:** Fix installed, not sim-tested yet. Before it, the centred version showed TA ONLY half outside its box, next to the FM message, in the sim.

What it is for

The TCAS message ("TA ONLY", "TCAS") sits at the left of its line at the bottom of the ND, as the A320 FCOM ND messages figure shows it (DSC-34-SURV-60-20, item 2: TA ONLY at the bottom, left of centre). The bottom centre stays free for the FM messages (GPS PRIMARY...). A first change tried to centre it with a CSS text-anchor plus a CSS transform; the sim renderer applies the text-anchor but not the transform on SVG text, so the message was centred on its left start point and stood half outside its box. That rule is removed. TCAS STBY is not shown on the A32NX ND (hidden by its stylesheet), as before.

How to use

1. Nothing to set: the message shows wherever the TCAS sends one.

How to test

#	Do this	Expected result
1	A32NX, ND in ROSE NAV or ARC. Set the TCAS mode selector on the ATC/TCAS panel to **TA**.	**TA ONLY** in white at the bottom of the ND, left of centre, fully inside its grey box; an FM message such as GPS PRIMARY, when shown, stays readable at the centre.
2	Set the TCAS mode selector back to **TA/RA**.	TA ONLY disappears.

8. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)

> **Aircraft:** A32NX | **Where:** Overhead GPWS panel (GPWS SYS, TERR, G/S MODE, FLAP MODE, LDG FLAP 3 pbs), GPWS/PULL UP lights on both glareshields, EMER CANC key, ND terrain display | **Status:** Tested in the sim by the user with the recorder (October 2026): take-off without false alert, FL100 toward Mount Fuji with the caution, the warning and AVOID TERRAIN; the ND TERR AHEAD message and threat areas tested the same way on 26 October 2026, and the AVOID TERRAIN voice the same day; the obstacle alerts at 3000 ft toward a 3125 ft tower near KFAR the same day (caution, warning, AVOID OBSTACLE). Not tested: the Himalaya at FL250, Innsbruck-type valley approaches. Unit tests for every part.

> **NOTE**
> This feature is in `core/fcom-gap-fixes` (commits 2eb8e9ab0, for the ND part 50e981ffb, for the AVOID TERRAIN voice e68a0e5c5, for the obstacle alerts 5ffa81a07), not in the branch of the rest of this manual: it needs the TERR STBY and terrain display work that is there. It is also on `develop`. The A380X GPWS is not changed.

What it is for

Before, the A32NX GPWS only knew the basic modes, which work from the radio altimeter. Flying toward a mountain at FL100 gave no alert until the ground rose under the aircraft, far too late. The GPWS now also looks ahead: it knows the terrain around the aircraft from an elevation grid, and warns when your flight path and the terrain meet. Modes 1 and 2 were written again at the same time.

What you see and hear

Situation	You hear	You see
Terrain ahead, you can still avoid it (caution)	"TERRAIN AHEAD", repeated every 7 s	Amber GPWS light on both GPWS/PULL UP pbs; the terrain comes up on both NDs by itself, with an amber **TERR AHEAD** and the threat cells in yellow
Terrain right below the aircraft, too low for the place (terrain clearance floor)	"TOO LOW TERRAIN", repeated every 7 s	Amber GPWS light; the terrain does not come up by itself and there is no ND message
Terrain ahead, close (warning)	"TERRAIN AHEAD" once, then "PULL UP" again and again	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the threat cells in red
Terrain ahead and the aircraft could not climb over it even with a pull-up	"AVOID TERRAIN" again and again (about every 1.5 s)	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the warning cells in a red and black crosshatch
An obstacle (tower, mast) ahead, you can still avoid it (caution)	"OBSTACLE AHEAD", repeated every 7 s	Amber GPWS light; the terrain comes up on both NDs with an amber **OBST AHEAD** and the threat cells in yellow
An obstacle ahead, close (warning)	"OBSTACLE AHEAD" once, then "PULL UP" again and again	Red PULL UP light; red **OBST AHEAD**, the threat cells in red
An obstacle ahead that a pull-up could not clear	"AVOID OBSTACLE" again and again (about every 1.5 s)	Red PULL UP light; red **OBST AHEAD**, the warning cells in the red and black crosshatch
Mode 1 second boundary (sink rate too high near the ground), or mode 2 warning	"PULL UP"	Red PULL UP light
Any other basic alert (SINK RATE, mode 2 caution TERRAIN, DON'T SINK, TOO LOW ..., GLIDESLOPE)	The matching words	Amber GPWS light

- A warning takes over from a caution. Once AVOID TERRAIN has been said, PULL UP does not come back in the same warning.
- Terrain comes first: when the same scan finds terrain and an obstacle, you get the terrain alert. The obstacle words only come when the threat is an obstacle alone.
- The terrain display comes up on both NDs and the TERR ON ND pbs light, whatever they were set to. During the alert each pilot can switch it off with the TERR ON ND pb.
- **The AVOID TERRAIN voice:** the sound bank has no such word, so "AVOID" was made with a speech synthesiser converted to the GPWS voice and joined to the "TERRAIN" word of the sound bank (`tools/gpws-voice/generate.py` rebuilds it). The GPWS says one word at a time: the last PULL UP ends before the first AVOID TERRAIN, they never overlap. "AVOID OBSTACLE" is made the same way and waits for the last OBSTACLE AHEAD or PULL UP in the same way.
- **TERR STBY:** the predictive functions need the GPS position (accurate to 0.5 NM or better), the GPS altitude, the vertical speed and the terrain grid around the aircraft. When one is missing for 3 s the terrain function goes to standby (TERR STBY on the ND, no TAD or TCF alerts); 1 s after everything is back it works again. A block of the grid that is merely slow to load does not make it flicker.

On the ND: TERR AHEAD and the threat areas

- **OBST AHEAD** takes the place of TERR AHEAD, in the same colours and with the same flashing, when the alert is about an obstacle. The obstacle cells are drawn in the threat areas like terrain cells.
- **TERR AHEAD** on both NDs: amber for the caution TERRAIN AHEAD, red for the warning (TERRAIN AHEAD PULL UP or AVOID TERRAIN). TOO LOW TERRAIN has no ND message (FCOM: nil). It takes the place of the cyan **TERR** label above the terrain elevation box.
- It flashes for 9 s each time it is triggered (again when the caution becomes a warning), then stays steady while the alert lasts.
- If the terrain picture cannot be shown, the ND says why in its centre, red or amber with the alert: **TERR: CHANGE MODE** in PLAN mode, **TERR: REDUCE RANGE** at 160 and 320 NM.
- **Threat areas** are drawn over the terrain picture: the 1 arcminute cells (about 1 NM) that break the caution envelope in solid yellow, those that break the warning envelope in solid red, and in a red and black crosshatch while AVOID TERRAIN is on.
- They come from the look-ahead strip along your track, which is narrow and widens in turns. So they are small patches where your path meets the terrain, not whole mountains. Up to 64 runs of cells are drawn; when there are more, the nearest ones are kept.



Figure 14. Caution (ARC, 20 NM): amber TERR AHEAD at the lower right, the threat cells in yellow on the track.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 15. Warning: red TERR AHEAD, the warning cells in red next to the yellow caution cells.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 16. AVOID TERRAIN: the warning cells in the red and black crosshatch, yellow caution cells beside them.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The functions in plain words

- **Terrain ahead alerting (TAD):** the GPWS scans the terrain along and beside your track, and checks whether you could clear it by levelling off and then climbing gently (caution), and whether you could still clear it if you pulled up now (warning). Both envelopes end on such a climb path, so the caution comes about a minute before the threat and the warning about half a minute before (the 3 degree climb of the caution path is a design choice; the manual gives no angle). It does not alert on the ground.
- **Obstacle alerts:** the same scan also reads an obstacle grid (the highest obstacle top in each cell) and gives the OBSTACLE words for it. No extra margin is added for obstacles, because their heights are surveyed values, not averaged terrain.
- **Terrain clearance floor (TCF):** around the destination runway (up to 15 NM) there is an invisible floor above the terrain. It starts low at the runway and rises with the distance. If you descend below it, you get TOO LOW TERRAIN whatever the gear and flaps are. The shape of the floor is rebuilt from the figures of the Honeywell guide; it is not a published curve.
- **Mountainous approach areas:** where the terrain within 6 NM of the runway is more than 2000 ft above it, the warning looks only 30 s ahead instead of 120 s, so that you are not warned on every approach there.
- **Landing tunnel:** when you are configured for landing (gear down, landing flaps), inside the approach sector of the runway (2.7 NM), in a tunnel above the approach path, and your track converges on the runway, the alerts are inhibited. That is how a normal landing near hills stays quiet.
- **Take-off:** no alerts on the ground; the terrain clearance floor does not work within 2.5 NM of the departure runway, and the scan starts afresh at lift-off.

The pushbuttons and keys

Control	What it does
TERR pb OFF (overhead)	Inhibits TAD and TCF. Mode 2 is then active again (it is switched off while the predictive functions work).

GPWS SYS pb OFF	Inhibits the basic modes 1 to 5 and their lights. TAD and TCF keep working: they do not need the radio altimeter or the airspeed.
EMER CANC	Silences the words of the alert that is going on (also the repeats), until that alert is over. The lights stay on. A new alert speaks again.
Audio inhibit	Stops the words of the predictive alerts too.

Modes 1 and 2 after the rewrite

- The lights follow the real format: **red PULL UP only for the second boundary of mode 1 and for a mode 2 warning** (and the hold after mode 2); every other alert is the amber GPWS light.
- Mode 2 has two nested envelopes: the outer gives the caution ("TERRAIN, TERRAIN"), the inner the warning ("PULL UP"). They depend on the radio altitude, its closure rate and the configuration.
- Mode 2 is inhibited while TAD and TCF work. It comes back when they fail, when TERR is OFF, or when the radio altitude and the height above the terrain computed from the grid differ by more than 2000 ft (one of the two is then wrong).
- The glideslope bias of mode 1 is applied only for a valid glideslope deviation. EMER CANC latches the silence until the alert is over.

The escape manoeuvre

On a PULL UP or AVOID TERRAIN warning: autopilot off, pitch up firmly, TOGA thrust, wings level, and keep climbing until the alert stops (the usual Airbus terrain escape; the GPWS does not fly it for you).

Limitations and why

Limitation	Why	What to do
Obstacle alerts in the United States only.	The obstacle data is the FAA Digital Obstacle File, which lists the obstacles of the United States and its territories. Elsewhere there is no obstacle alert (and no TERR STBY for it: the terrain functions do not need the obstacle file).	Outside the United States, do not rely on it for man-made obstacles.
The obstacle data is a snapshot.	The grid is built once from the FAA file (26 October 2026); obstacles built or removed later are not in it until the file is rebuilt.	-
An obstacle alert can come a few seconds early.	An obstacle fills its whole 1 arcminute cell (about 1 NM), so the scan meets it up to about 1 NM before the real tower.	-
No runway database.	TCF, the mountainous approach area and the landing tunnel only know the runway threshold the FMGC sends for its destination and the place you took off from (the departure runway). An airport that is neither is not protected by a floor.	Enter the destination in the flight plan, with its runway when you know it.
Valley airports can alert.	The grid cells are 1 arcminute (about 1 NM), so a narrow valley airport such as Innsbruck sits in cells with the heights of the surrounding mountains. A margin that grows with the relief around the cell is added because the source data is averaged and reads peaks too low.	Expect TOO LOW TERRAIN or TERRAIN AHEAD there; set TERR OFF if you must.
Coverage from latitude -84 to 83.	The terrain source has nothing beyond that.	Beyond it: TERR STBY.

The grid is a file, and without it there is TERR STBY.	The grid is `panel/terrain_grid.bin` in the A32NX package folder (about 20 MB). It is too big for the repository, so it lives in the large-files repository (`TCdr/aircraft-large-files`, branch `feat/a32nx-taws-terrain-grid`). An aircraft without it has no predictive terrain alerts and shows TERR STBY.	Install a build that contains the file.
Threat areas only where the terrain picture is shown.	They are drawn over the terrain picture and cut to its area, so there are none in PLAN mode (TERR: CHANGE MODE instead).	Use ARC or ROSE NAV.
Threat areas are 1 arcminute cells.	That is the resolution of the terrain grid (about 1 NM): the patches are blocky and show the cells of the look-ahead strip, not the shape of the mountain.	-
AVOID TERRAIN is a made-up recording.	The sound bank has no such word: "AVOID" is synthesised and converted to the GPWS voice, so it can sound slightly different from the recorded words.	-
Position accuracy of 0.5 NM or better is needed.	Design choice: the accuracy at which the real system deactivates is not published. Half a cell was chosen so that the cell under the aircraft is the cell read.	-

Where the data comes from

The elevation grid is derived from the terrain data of SimBridge (the data behind the ND terrain display), reduced to one byte per 1 arcminute cell: 0 is water, otherwise the highest elevation of the cell in 50 m steps. It is built by `tools/taws-terrain-grid/build_grid.py`. The source is already averaged (Fuji reads about 3650 m, the real summit is 3776 m), which is why the alert logic adds its own margins. The aircraft reads the file from its own package while you fly; SimBridge does not have to run and nothing goes to the internet.

The obstacle grid `panel/obstacle_grid.bin` (about 265 KB, in the repository) has the same format: in each cell the highest obstacle top above sea level, rounded up to 50 m. It is built from the FAA Digital Obstacle File by `tools/taws-obstacle-grid/build_obstacle_grid.py`.

How to use

1. Nothing to set. Fly with the GPS working and a destination in the flight plan; the GPWS SYS and TERR pbs stay in.
2. Set TERR to OFF only to inhibit TAD and TCF (for instance on a deliberately low valley approach). Press EMER CANC to silence a repeating alert once you have taken note of it.

How to test

#	Do this	Expected result
1	A32NX in flight, GPS aligned, TERR and GPWS SYS pbs in. Look at the ND.	No TERR STBY. (Without `terrain_grid.bin` in the panel folder: TERR STBY.)
2	Take off normally from an airport near hills.	No alert on the ground or at lift-off.
3	Cruise at FL100 on a track that meets high ground (for example RJAA to RJOA, toward Mount Fuji).	About a minute before the terrain: "TOO LOW TERRAIN" then "TERRAIN AHEAD" every 7 s, amber GPWS light, terrain on both NDs.

4	Keep the course.	The warning: "TERRAIN AHEAD" once, "PULL UP" repeating, red PULL UP light. When a pull-up could not clear it: "AVOID TERRAIN", about every 1.5 s, never over a PULL UP.
5	Look at both NDs in ARC at 20 NM during the caution, then the warning, then AVOID TERRAIN.	Caution: amber TERR AHEAD in place of the cyan TERR label, flashing 9 s then steady, yellow cells on the track. Warning: TERR AHEAD red and flashing again, red cells. AVOID TERRAIN: the red cells crosshatched red and black.
6	During the alert, select PLAN mode, then ARC at 160 NM.	TERR: CHANGE MODE, then TERR: REDUCE RANGE, in the ND centre, in the colour of the alert; no threat areas.
7	Press EMER CANC during the alert.	The words stop and do not repeat for that alert; the lights stay.
8	Turn away and climb.	The alert ends, the lights go off. A new threat speaks again.
9	In the United States, fly at 3000 ft and 260 kt straight at a tall tower (for example from KFAR, DIR TO 4716.7N/09720.5W, a 3125 ft tower) and keep the level.	About 75 s out: "OBSTACLE AHEAD" every 7 s and amber OBST AHEAD; about 30 s out: "OBSTACLE AHEAD", "PULL UP", red OBST AHEAD; then "AVOID OBSTACLE" about every 1.5 s, never over a PULL UP.
10	Climb (AP off, pitch up, TOGA).	The obstacle alert ends.
11	Set the overhead TERR pb OFF in front of terrain.	No TAD or TCF alert; mode 2 works again from the radio altimeter.

9. TCAS traffic display, ND TCAS messages, TRAF pb, WXR/TAWS SYS pbs

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display and messages; A32NX TRAFFIC selector; A380X FCU TRAF pb, pedestal SURV panel, MFD SURV STATUS & SWITCHING | **Status:** Not sim-tested (added, unit tests only: TcasTrafficSelector, TcasSettingMessage, SurvWxrTawsSysPbs and TcasTrafAutoSelect specs, all failing without the change).

What it is for

Deviations from the FCOM TCAS and SURV descriptions found by the FCOM gap audit (A320 FCOM DSC-34-SURV-40 TCAS, A380 FCOM SURV TCAS and SURV panel descriptions):

- **A32NX TRAFFIC selector THRT:** while a TA or an RA is present, the proximate and the other intruders within ±2700 ft are shown too, not only the threat (A320 FCOM: "THRT: Proximate and other intruders are displayed only if a TA or an RA is present").
- **ND TCAS messages (both aircraft): TCAS: CHANGE MODE** in PLAN mode and **TCAS: REDUCE RANGE** above the 40 NM range while there is a TA or an RA; the message is amber for a TA and red for an RA.
- **A380X only: TCAS: INCREASE RANGE** in ROSE VOR or ROSE LS with the ZOOM range in flight; no such message in ZOOM on the ground.
- **A380X WXR/TAWS SYS 1 and SYS 2 pushbuttons** (pedestal SURV panel and MFD SURV STATUS & SWITCHING): pressing the pushbutton that is already lit keeps that system selected. Before, it selected no system.
- **A380X TRAF pb:** the TCAS now reads the FCU TRAF light (the variable it read before was written by nothing). A TA or an RA selects TRAF on each side once, so the traffic shows; the crew can then deselect it.

Situation	What you should see
TA or RA present, ND in PLAN mode	ND message **TCAS: CHANGE MODE** (amber for a TA, red for an RA).
TA or RA present, ND range above 40 NM	ND message **TCAS: REDUCE RANGE** (amber / red).

A380X, TA or RA present, ROSE VOR or ROSE LS in ZOOM, in flight	ND message **TCAS: INCREASE RANGE**. On the ground in ZOOM: no message.
A380X, TRAF off on the FCU, a TA or an RA starts	TRAF comes on by itself on both sides (once); a press turns it off again.
A32NX, TRAFFIC selector THRT, a TA or an RA	The threat plus the proximate and other intruders within ±2700 ft show on the ND; with no TA/RA, no traffic.

Limitations and why

Limitation	Why	What to do
No TCAS TEST.	Not done yet.	-
No-bearing intruder lines are not shown.	Not done yet.	-

How to use

1. Nothing to set: the messages and the TRAF selection follow the TCAS.
2. A32NX: use the THRT position of the TRAFFIC selector (ATC/TCAS panel) to see traffic only around a TA or an RA.
3. A380X: select the WXR/TAWS system with its SYS pushbutton; pressing the lit one again changes nothing.

How to test

#	Do this	Expected result
1	In flight with traffic (AI or multiplayer) that gives a TA. ND in ARC, range 80 NM.	ND: **TCAS: REDUCE RANGE** in amber (red during an RA).
2	During the TA, select PLAN mode.	ND: **TCAS: CHANGE MODE**.
3	A380X: TRAF off on the FCU before the TA.	At the TA the TRAF light comes on, on both sides; press TRAF: it goes off and stays off for that TA.
4	A380X: ROSE LS, ZOOM range, in flight, during a TA.	ND: **TCAS: INCREASE RANGE**.
5	A380X on the ground: MFD > SURV > STATUS & SWITCHING, WXR/TAWS SYS 1 lit. Press SYS 1 again (MFD page, then the pedestal pushbutton).	SYS 1 stays selected and lit (before: no system).
6	A32NX: TRAFFIC selector THRT, traffic around but no TA.	No traffic on the ND; when a TA comes, the TA and the other intruders within ±2700 ft show.

What PR testers should check

- The ND message position: centred, above the MODE/RANGE CHANGE line.
- A380X: a TA turns TRAF on, and a second press turns it off.
- The TCAS message colours (amber TA, red RA) and the THRT traffic display on the A32NX.
- These changes are not sim-tested: report any message that does not match the FCOM.